

Nadim Mahmud

Machine Learning Engineer | Deep Learning, Computer Vision, Explainable AI

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SUMMARY

Machine learning engineer in the final semester of a B.Sc. in Software Engineering with a data science major (CGPA 3.73/4.00). I train and evaluate deep learning models for image classification and segmentation, and gradient boosting models for tabular data, in Python with PyTorch and scikit-learn. My projects focus on what happens after the first good score: leakage-safe data splits, stress tests with corrupted inputs, calibration checks and SHAP or Grad-CAM explanations. Three research papers submitted in 2026.

RESEARCH INTERESTS

Robust computer vision, explainable AI, image segmentation, machine learning for agriculture and the environment, causal machine learning.

EDUCATION

Daffodil International University

B.Sc. in Software Engineering, major in Data Science

Dhaka, Bangladesh

Expected 2026

- CGPA: 3.73 / 4.00
- Thesis: causal uplift modelling for email marketing (see Projects)

TECHNICAL SKILLS

Programming: Python, NumPy, Pandas, Jupyter, Git, GitHub, Google Colab, Kaggle

Machine learning: scikit-learn, XGBoost, HistGradientBoosting, Random Forest, Logistic Regression, feature engineering, cross-validation, hyperparameter tuning

Deep learning: PyTorch, torchvision, CNNs, transfer learning, fine-tuning, ResNet, DenseNet, EfficientNet, U-Net, Attention U-Net, data augmentation

Computer vision: image classification, semantic segmentation, robustness benchmarking under noise, blur and brightness changes

Explainability and evaluation: SHAP, Grad-CAM, permutation importance, probability calibration (ECE, Brier score), Dice, IoU, F1, ROC-AUC

Causal ML: uplift modelling, causal inference, confounding analysis

Visualisation: Matplotlib, Seaborn

EXPERIENCE

Machine Learning Intern

FlyRank AI

Jun 2026 – Present

Remote

- Prepare and clean datasets for machine learning experiments.
- Evaluate models, compare metrics across runs and write up the results for the team.

TQ-Tech (startup)

Product Manager Intern, Dec 2025 – Mar 2026; Junior Business Development Executive, Apr 2026

Dec 2025 – Apr 2026

Remote

- Wrote product requirements and feature specifications for the development team.
- Ran user and market research, including competitor analysis, to inform product decisions.
- Moved into a junior business development role after the internship and supported the team's business development work.

PROJECTS

IrrigaSmart: 12-hour soil-moisture risk prediction

Python, scikit-learn

github.com/nadim7009/irrigasmart-soil-moisture-risk

- Built an end-to-end pipeline on 31,492 sensor observations from vegetable farms in Belgium: removed duplicate and invalid readings, rebuilt crop-cycle boundaries and engineered 70 features from soil moisture, weather and field data.
- Split the data by whole crop cycle so no field-season appeared in both training and test sets, then compared five models against a simple threshold rule.
- HistGradientBoosting reached F1 0.977 and ROC-AUC 0.999 on unseen crop cycles and missed 44 at-risk cases against 81 for the rule baseline.

- Checked calibration (Brier score 0.0038) and corrupted up to 30% of sensor readings in stress tests; F1 dropped by less than 0.003.

Robust plant disease classification with corruption-aware training

PyTorch

- Fine-tuned EfficientNetV2-S, ResNet50, DenseNet121 and EfficientNet-B0 on PlantVillage (54,305 images, 38 classes) with a group-aware split that keeps photos of the same leaf together.
- Tested every model on nine corrupted copies of the test set (noise, low brightness and blur at three severity levels) and measured how far accuracy falls.
- Trained ResNet50 with blur, noise and brightness corruptions and compared it with standard training on both clean and corrupted images.
- Used Grad-CAM to check that predictions came from lesion areas and reported calibration with ECE and Brier score.

Microplastic segmentation in SEM images

PyTorch

- Trained a U-Net and a four-gate Attention U-Net on 237 labelled scanning electron microscopy images with BCE and Dice loss.
- Compared the two models on Dice and IoU, removed each attention gate in turn in an ablation, and tested the models under noise, brightness and contrast changes.

Explainable carbon-footprint prediction

XGBoost, SHAP

- Compared Random Forest, XGBoost and an MLP on 18,000 records with 49 behavioural features, using a scikit-learn pipeline that fits imputation, outlier clipping and encoding on training folds only.
- XGBoost gave the best test results of the three; SHAP values showed which lifestyle and digital-use features drove each prediction.

Undergraduate thesis: causal uplift modelling for email marketing

Causal ML

- Compare predictive and causal uplift models on the Hillstrom email campaign dataset with injected confounding, to show when a model that predicts response well still targets the wrong customers.

PUBLICATIONS (UNDER REVIEW)

"Beyond clean accuracy: controlled robustness evaluation and corruption-aware training for plant disease classification." N. Mahmud, M. S. Hossain, M. H. Riyad, J. J. Bushra, J. J. A. M. Sayer. Submitted to International Journal of Information Technology and Computer Science (IJITCS), MECS Press. Under review.

"Attention-enhanced deep learning for microplastic segmentation in scanning electron microscopy images." N. Mahmud and co-authors. Submitted to the International Conference on Advances in Artificial Intelligence (AAI 2026), UniMAP, Malaysia. Under review.

"Explainable machine learning for individual carbon-footprint prediction from traditional, digital and AI behaviours." N. Mahmud and co-authors. Submitted to an international conference. Under review.

CERTIFICATIONS

Data Science and Machine Learning with Python for Beginners, Ostad

LEADERSHIP

Media Executive, DIU Software Engineering Club

Oct 2023 – Jan 2026

- Ran media and public relations for club events; volunteered at the International Robotics Olympiad and Prothom Alo Math Olympiad.